

Yiwei Dong

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EDUCATION

Shenzhen University

Sep. 2020 – June 2024

Bachelor of Science in Information and Computing Science

- **GPA:** 85.1/100 (Top 15%)
- **Core Courses:** Mathematical analysis (A), Advanced Algebra (A), Probability theory (A), Numerical Analysis (A), Data Structures and Algorithms (A), Operational Research (A), Machine Learning (A)

HONORS AND SCHOLARSHIPS

Outstanding Graduate of College (Top 5%)	2024
Tencent Friend Scholarship (Top 5%)	2023
University-Level Third Price Scholarship (Top 10%)	2023
Ministry of Education - Huawei Future Star Scholarship (Top 1%)	2022 - 2023

RESEARCH EXPERIENCES

Research on Sales Forecasting Method based on Multimodal Data

Jan. 2024 – Aug. 2024

First Author

- Proposed a multimodal information fusion model, MFMA (Multimodal Fusion Model based on Attention), which utilizes the Encoder-Decoder architecture combining Multi-head Attention and LSTM
- Developed the Chinese Multimodal Movie (CMM) dataset, which includes 921 movies released from January 2017 to March 2024. Each movie in the dataset features a poster, three stills, and basic information about the film, with a total of over 500,000 review comments
- Completed a research paper *New Product Sales Time-series Forecasting Based on Multimodal Information Fusion Network* under the guidance of Dr. Wang Baohua

Research on Digital Application of Ancient Music

June 2023 – June 2024

Core Member

- Utilized multi-level Daubechies wavelet decomposition to effectively separate noise from signals
- Implemented thresholding techniques to accurately remove noise components while preserving the characteristics of the original audio, and applied wavelet reconstruction methods to synthesize the denoised audio signals
- The project was selected for the *Undergraduate Innovation and Entrepreneurship Training Program (Guangdong Province)* and was successfully concluded

COMPETITIONS

China Undergraduate Mathematical Contest in Modeling

Sep. 2022

Team Leader (Provincial Second Prize, Top 15%)

- Used Chi-square test and independent samples T-test to reveal significant differences between the weathering conditions and glass types and patterns, with lead-barium glass being more susceptible to weathering
- Developed a binary logistic regression model to classify high-potassium and high-lead-barium glasses, and used the class-average method for subclassification
- Used Spearman's rank correlation coefficient to analyze the correlation of chemical composition

China Artificial Intelligence Innovation Application Competition

Feb. 2023

Core Member (Third Prize, Top 10%)

- Completed the *Tuberculosis Auxiliary Diagnosis* project
- Developed the YOLO-Dense model by combining the input layer of YOLOV5 with the attention sub-module of the DenseNet model, which is used for identifying tuberculosis lesions in X-ray images
- The model achieved a nodule detection rate of 94.3% and an accuracy of 83.7%

Shenzhen, Hong Kong and Macau Undergraduate Maker Competition

Dec. 2021

Core Member (Excellence Award, Top 4)

- Completed the *Bionic Caterpillar Telescopic Coconut-picking Robot* project
- Used Pandas and NumPy to process data from sensors on the robot's limbs and employed LSTM to build an action prediction model for early motion forecasting of the robot
- **The project also won the third prize in the "Liyuan Challenge" Innovation and Entrepreneurship Competition at Shenzhen University**

EXTRACURRICULAR ACTIVITIES

Huawei Intelligent Base Club, SZU

June 2022 – June 2023

President

- Communicated with the university and Huawei company, recruited new members, organized science and technology lectures and led the team to participate in science and technology competitions
- Organized more than ten lectures, and led the club to win four national awards and nine provincial awards
- Awarded the Tencent "Contributors to the Cultural Inheritance of Student Associations" award by SZU

Ministry of Education Higher Education School Teacher-Student Exchange Program of Hong Kong, Macau and Mainland China

July 2021

Participator

- Participated in an exchange program and training camp (*Pilot Elite Program Training Camp of Shenzhen University and University of Macau*) and completed innovation and entrepreneurship courses
- Won the first prize in the competition at the end of the program and training camp

TECHNICAL SKILLS

- **Languages:** English (Proficient), Mandarin (Native)
- **Certifications:** Tencent Cloud practitioner (Mar. 2022 – Mar. 2024)
- **Skills:** Python (Proficient in Pytorch, Pandas and Numpy), Matlab (Intermediate), C++ (Intermediate)